POLITECNICO
MILANO 1863

Geoinformatics Project

Topic: Implementation of Long Short-Term Memory (LSTM) Models for the Prediction of Vegetation Indices (NDVI, NDMI) from Time Series of Meteorological Variables for Lombardy Region, Italy

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Introduction

- Vegetation indices like NDVI and NDMI are vital for ecological and agricultural monitoring but are limited by satellite cloud cover, causing data gaps (Rouse et al., 1974; Gao, 1996; Wilson and Sader, 2002).
- Traditional gap-filling methods are effective only for short gaps, while physically-based and machine learning models face challenges in complex landscapes and long-term dependencies (Michishita et al., 2014; Farbo et al., 2024).
- Deep learning architectures, particularly LSTM, show promise for accurately reconstructing vegetation indices by modeling temporal dynamics, but their performance in heterogeneous terrain needs further evaluation (Hochreiter and Schmidhuber, 1997; Guo et al., 2024).

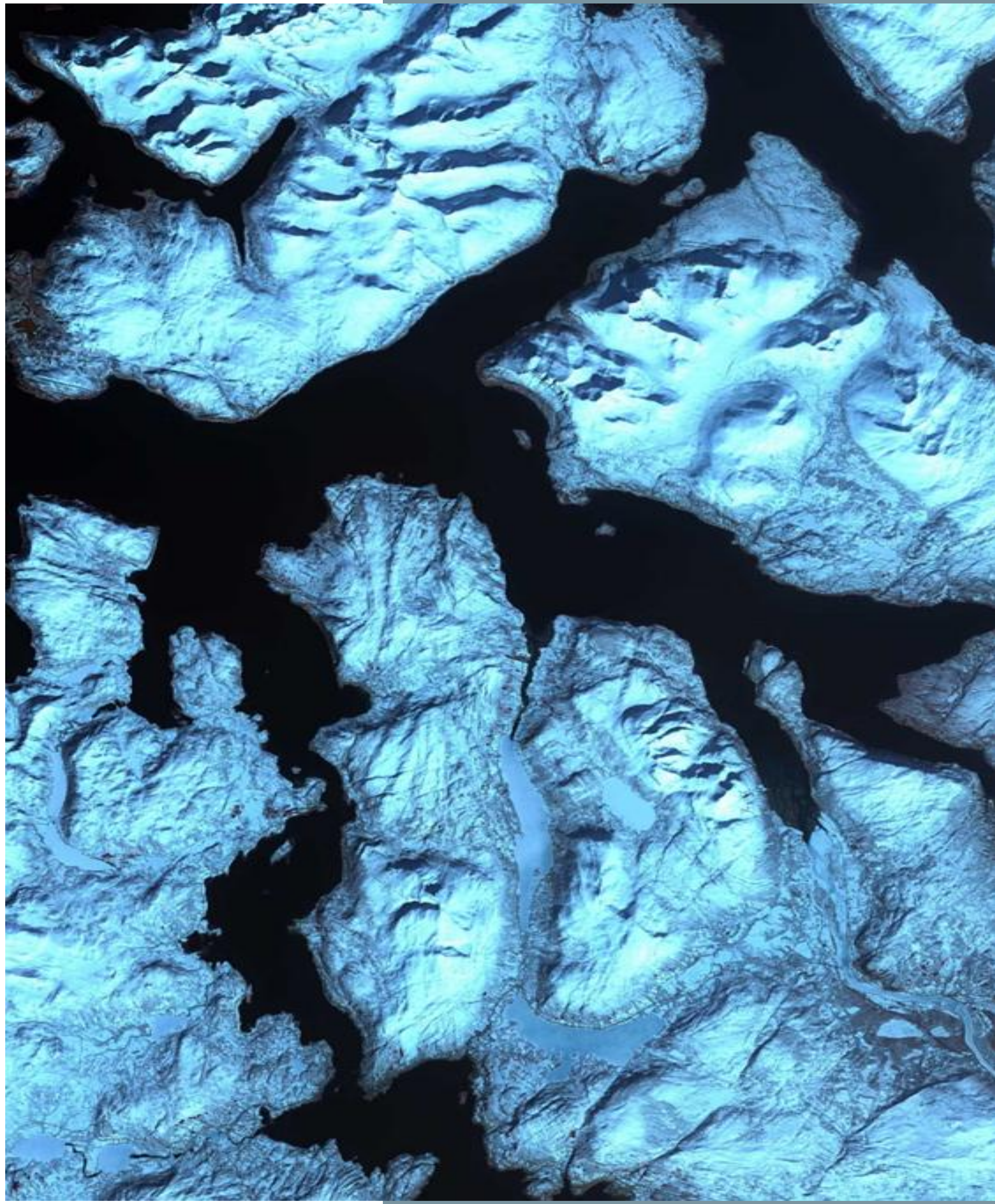
Study Aim and Objectives

Aim:

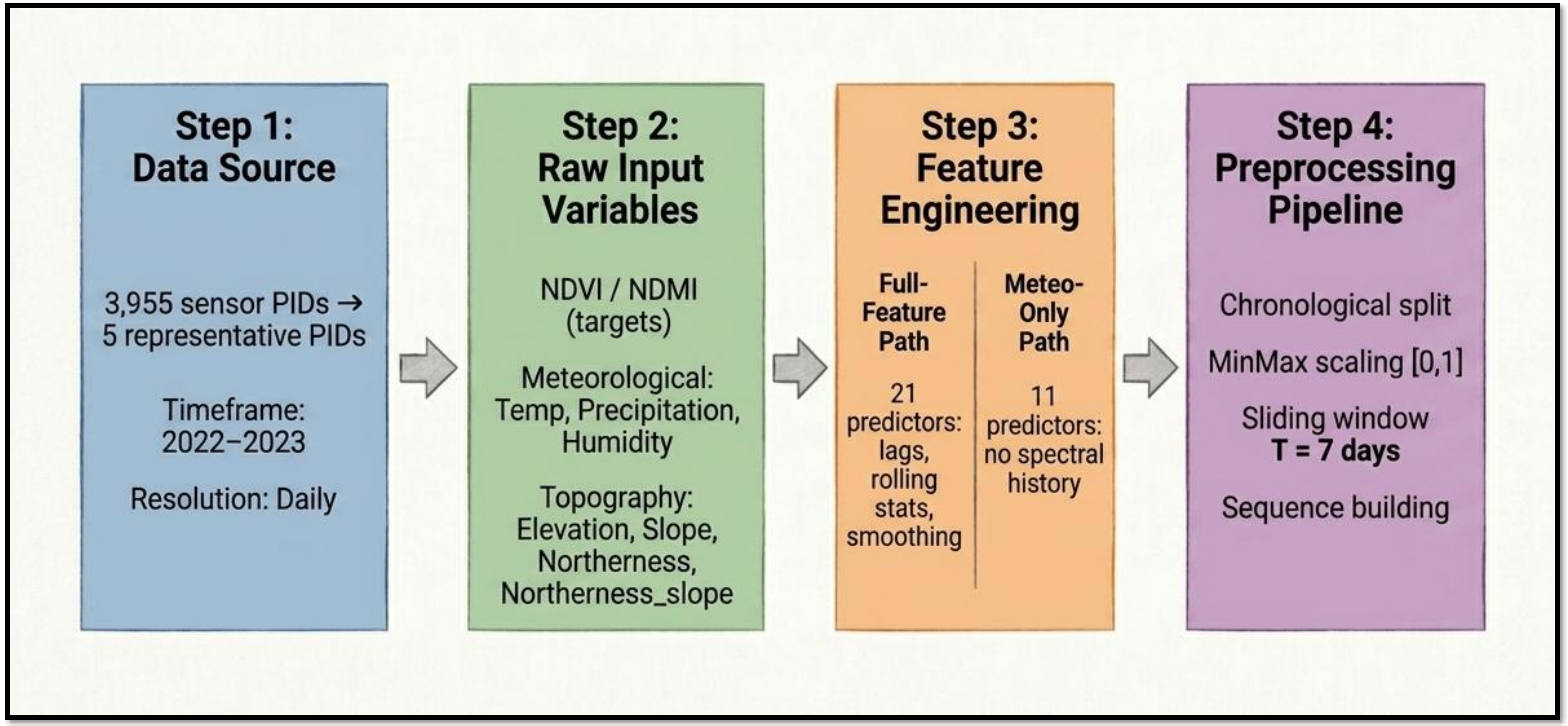
- To systematically evaluate and compare the performance of four LSTM-based deep learning architectures for simultaneous gap-filling and prediction of NDVI and NDMI vegetation indices in the complex and heterogeneous landscapes of the Lombardy region, Italy.

Objectives:

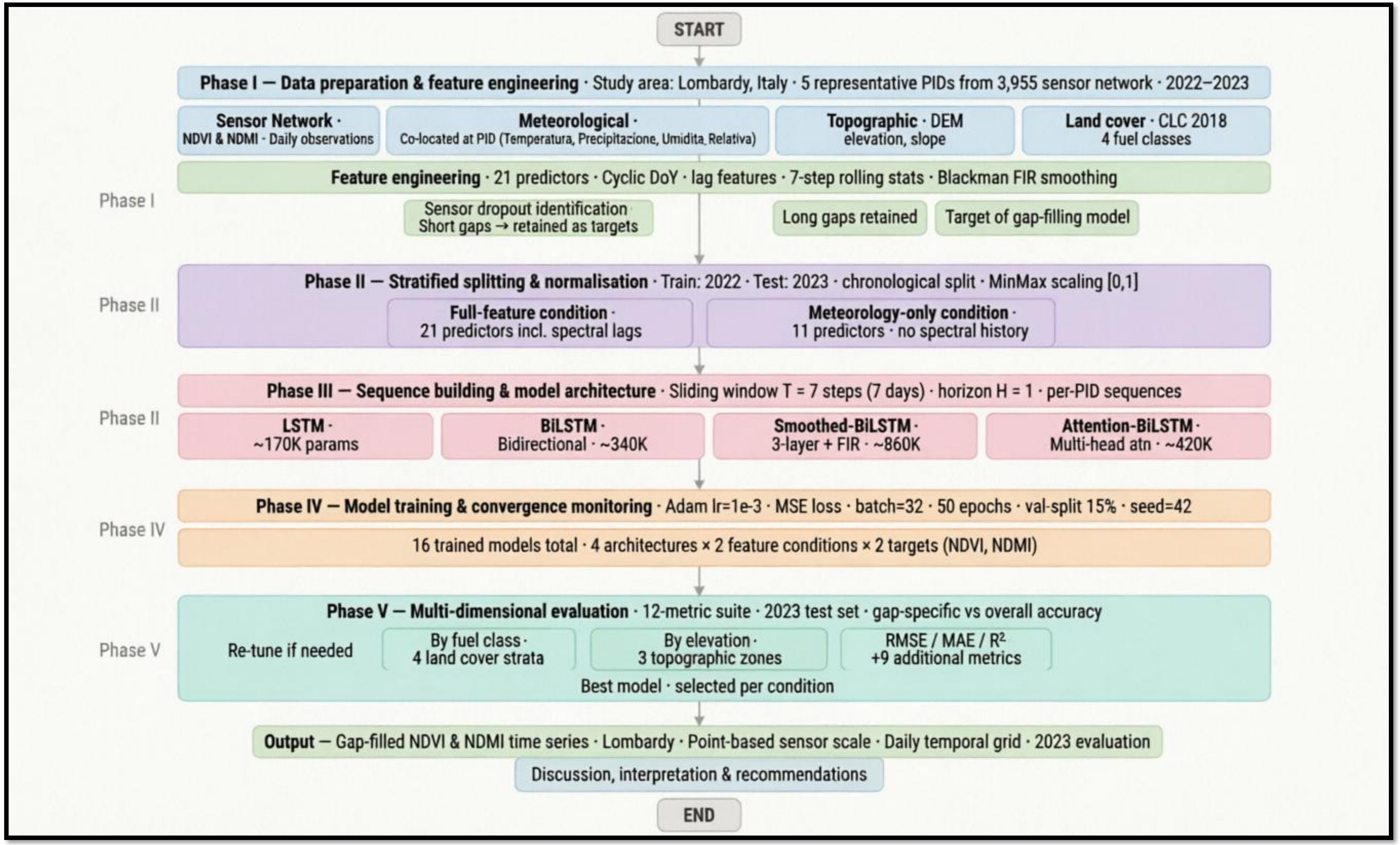
- Implement and compare four LSTM architectures (stacked LSTM, Bidirectional LSTM, Smoothed-BiLSTM, Attention-BiLSTM) for NDVI and NDMI gap-filling using multispectral satellite and meteorological data.
- Quantify the contribution of spectral index history versus meteorological inputs in model performance through ablation experiments.
- Assess model accuracy across different land cover types, topographic features, and seasonal phenological phases to identify spatial and temporal variability in prediction reliability.



Input Data Preparation



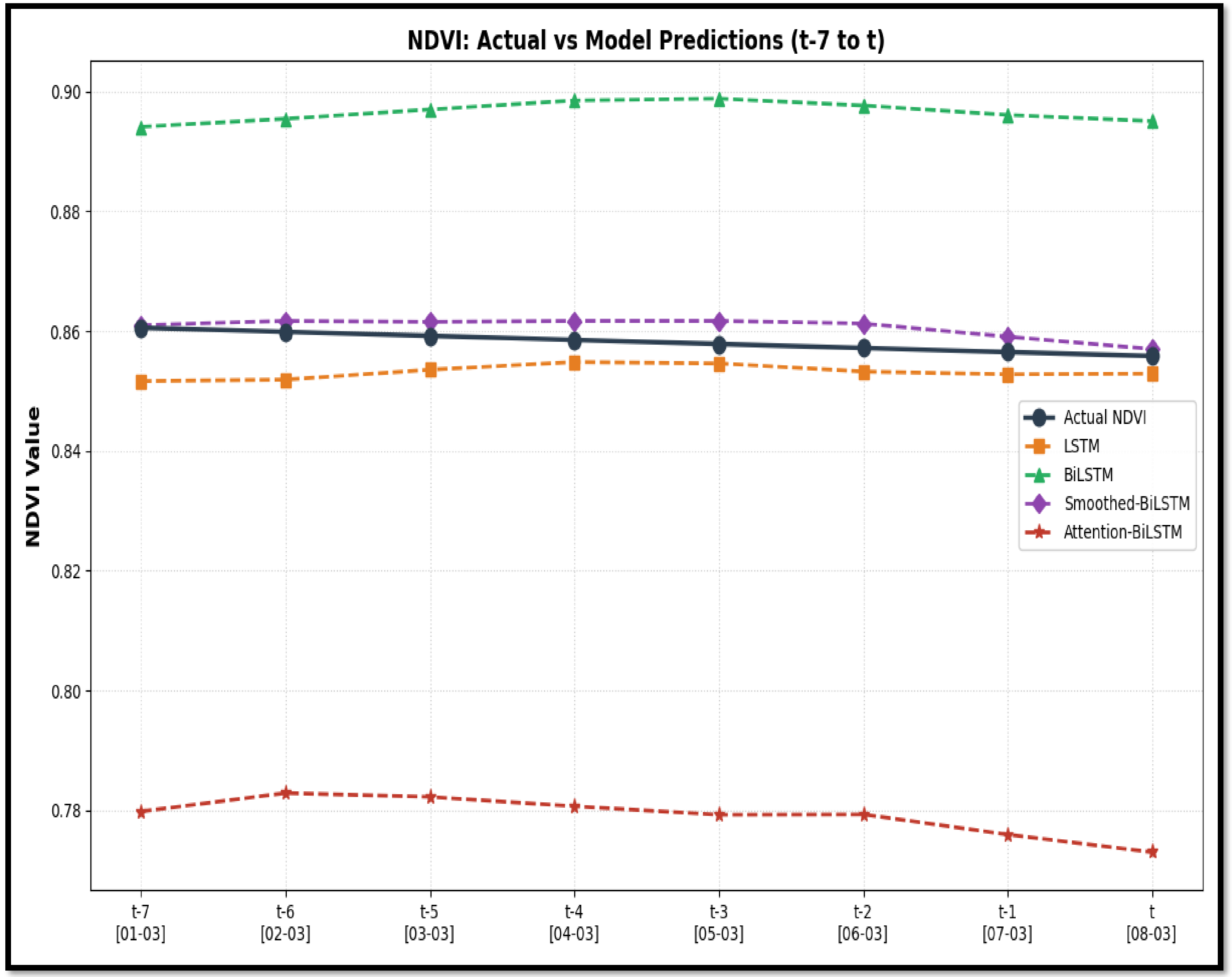
Methodological Flowchart

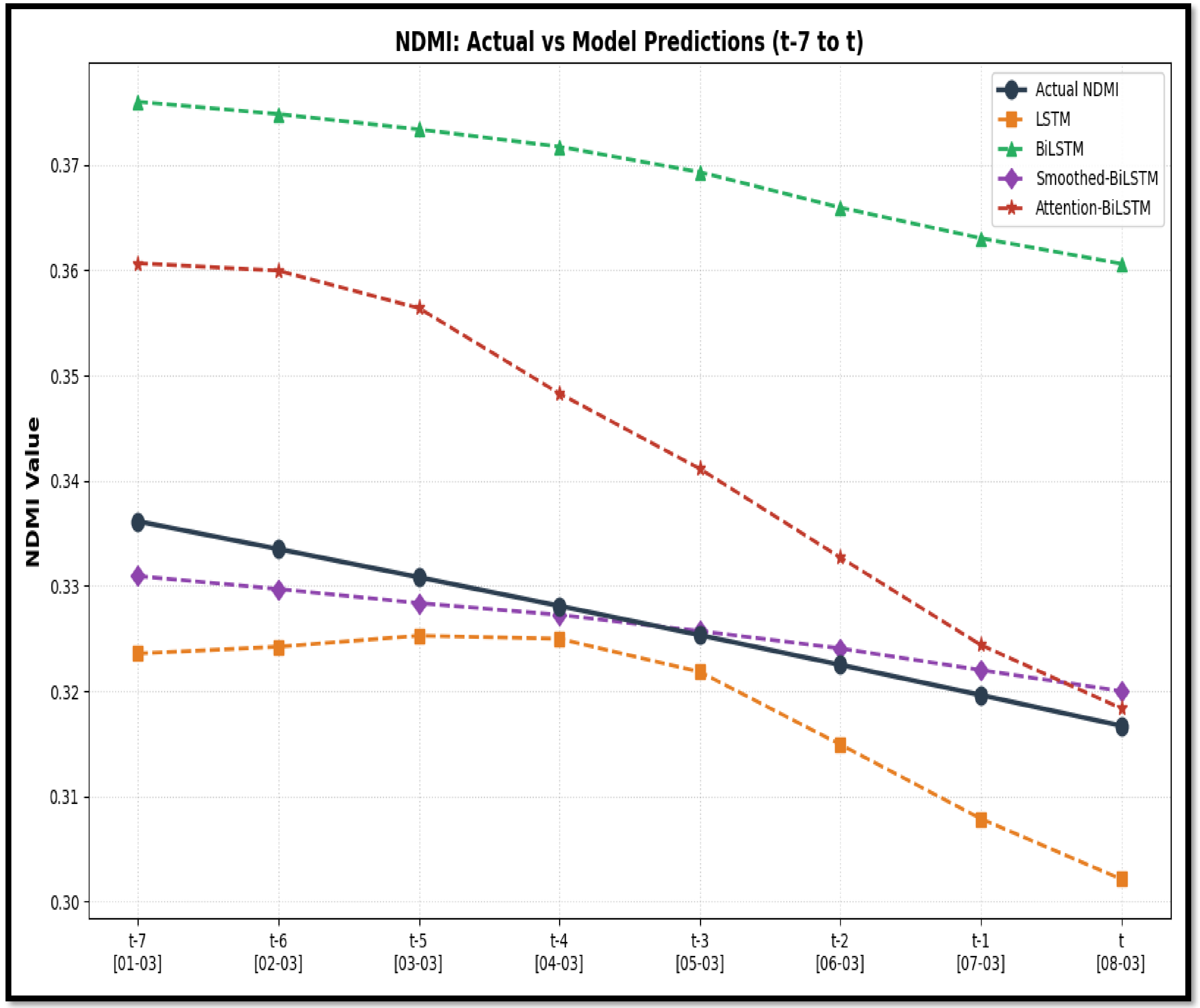


Softwares & Tools Used: Visual Studio Code, R, Python libraries

Working Model

7 Timestep Sample Sliding Window



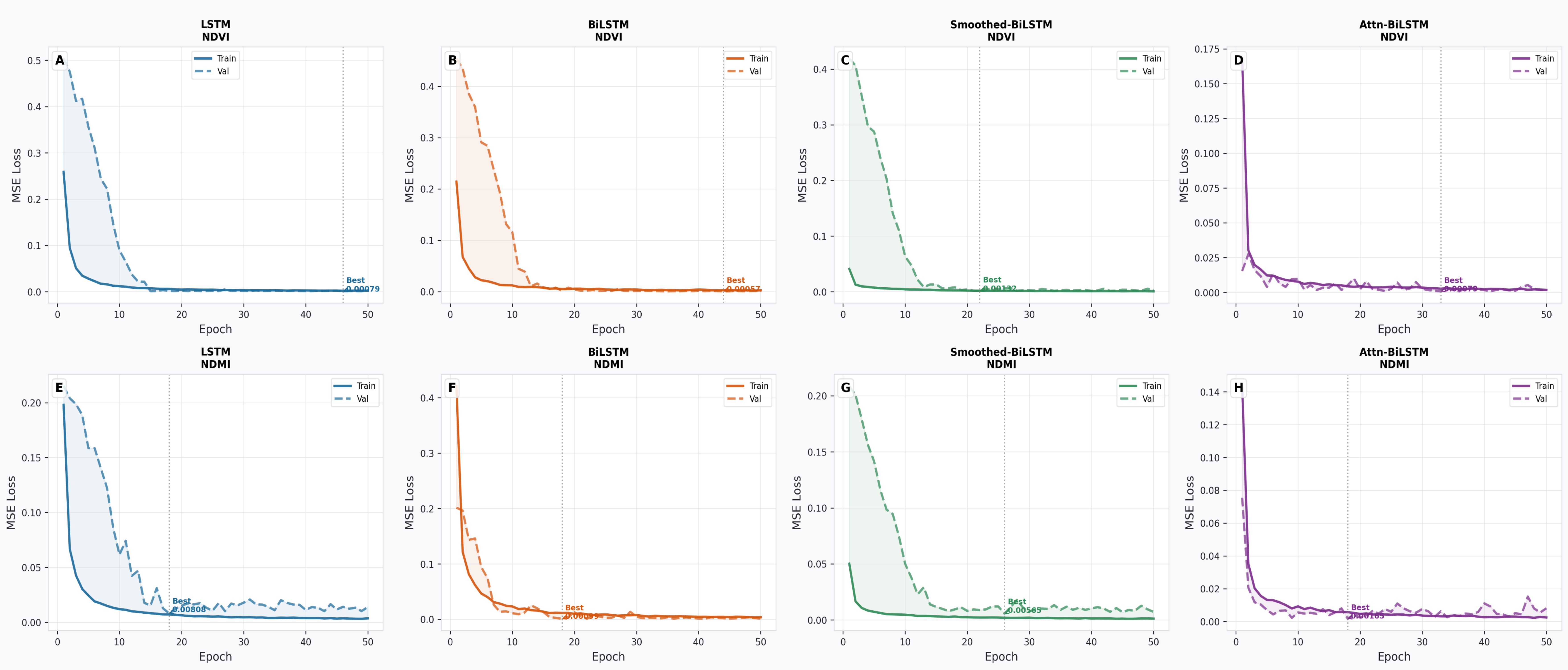


- The target value y_t is predicted using a sliding window of the previous 7 observations $(y_{t-1}, \ldots, y_{t-7})$ as input to the model.

- Each prediction at time t is a one-step-ahead forecast, where $y_t = f(y_{t-1}, \ldots, y_{t-7})$, and is compared with the actual value to assess error.

Data Analysis

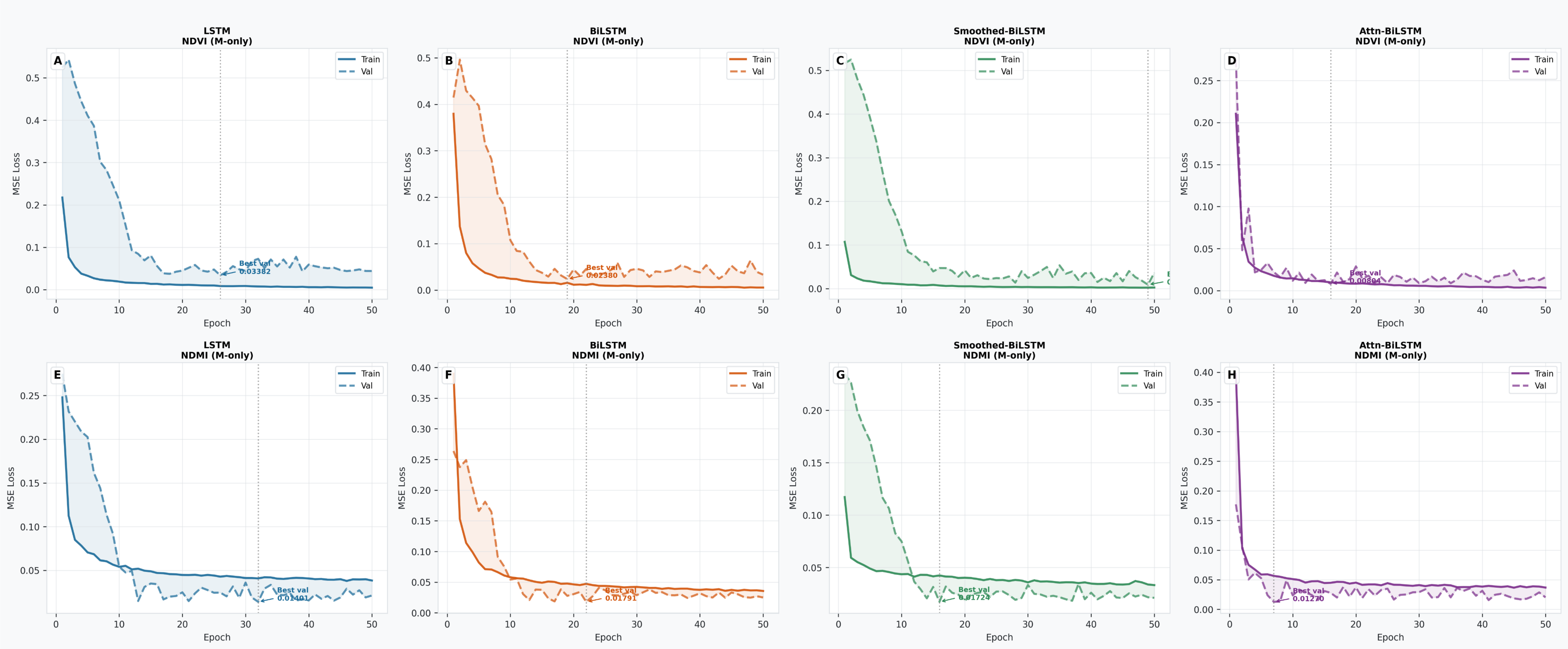
Model Training Convergence



- Rapid convergence within 10 epochs; LSTM achieved lowest validation MSE for NDVI, BiLSTM showed oscillations; all models stabilized before epoch 30 for NDMI, ensuring effective training without overfitting

Data Analysis

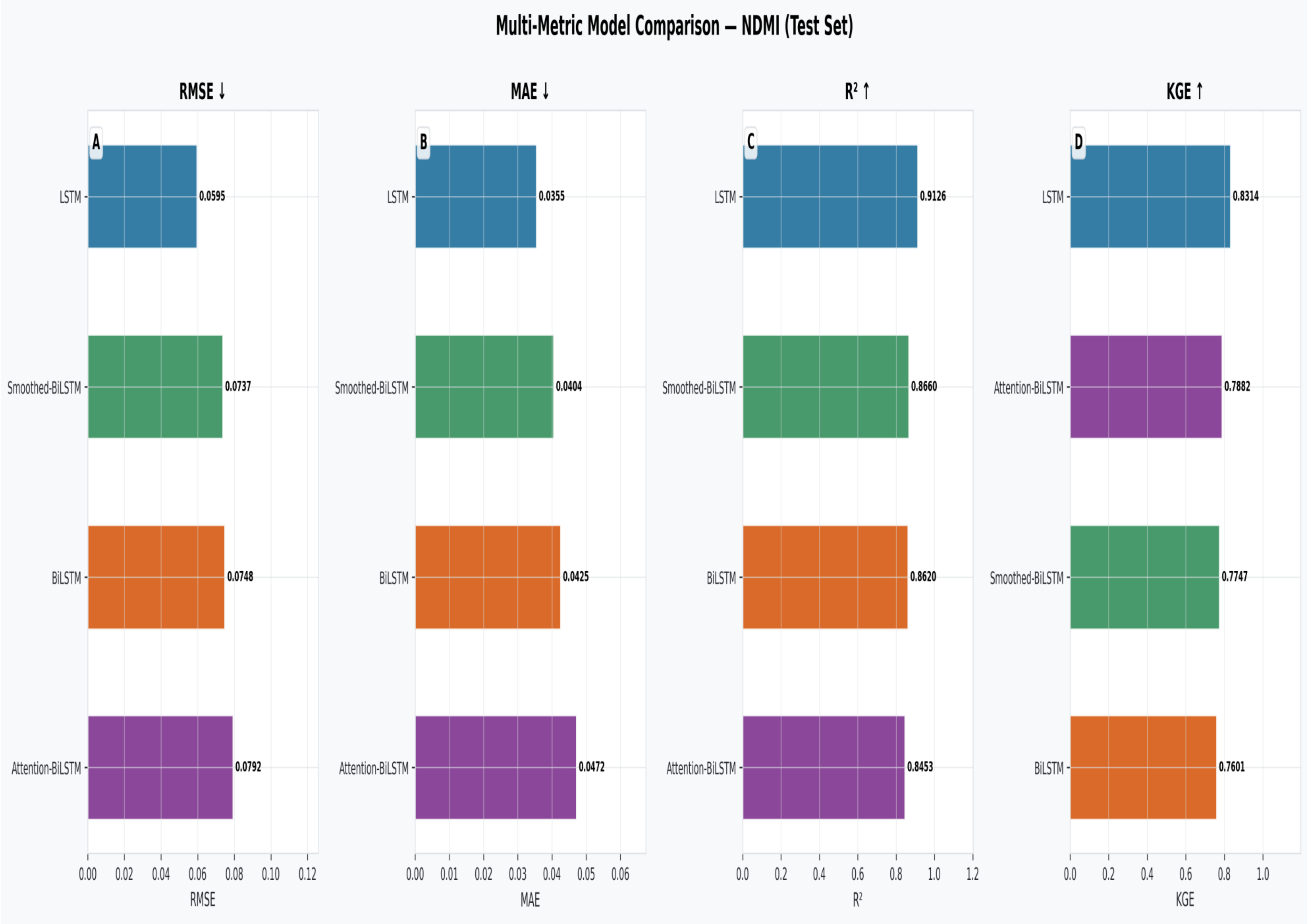
Meteo-Only Baseline: The Role of Spectral History as a Predictor

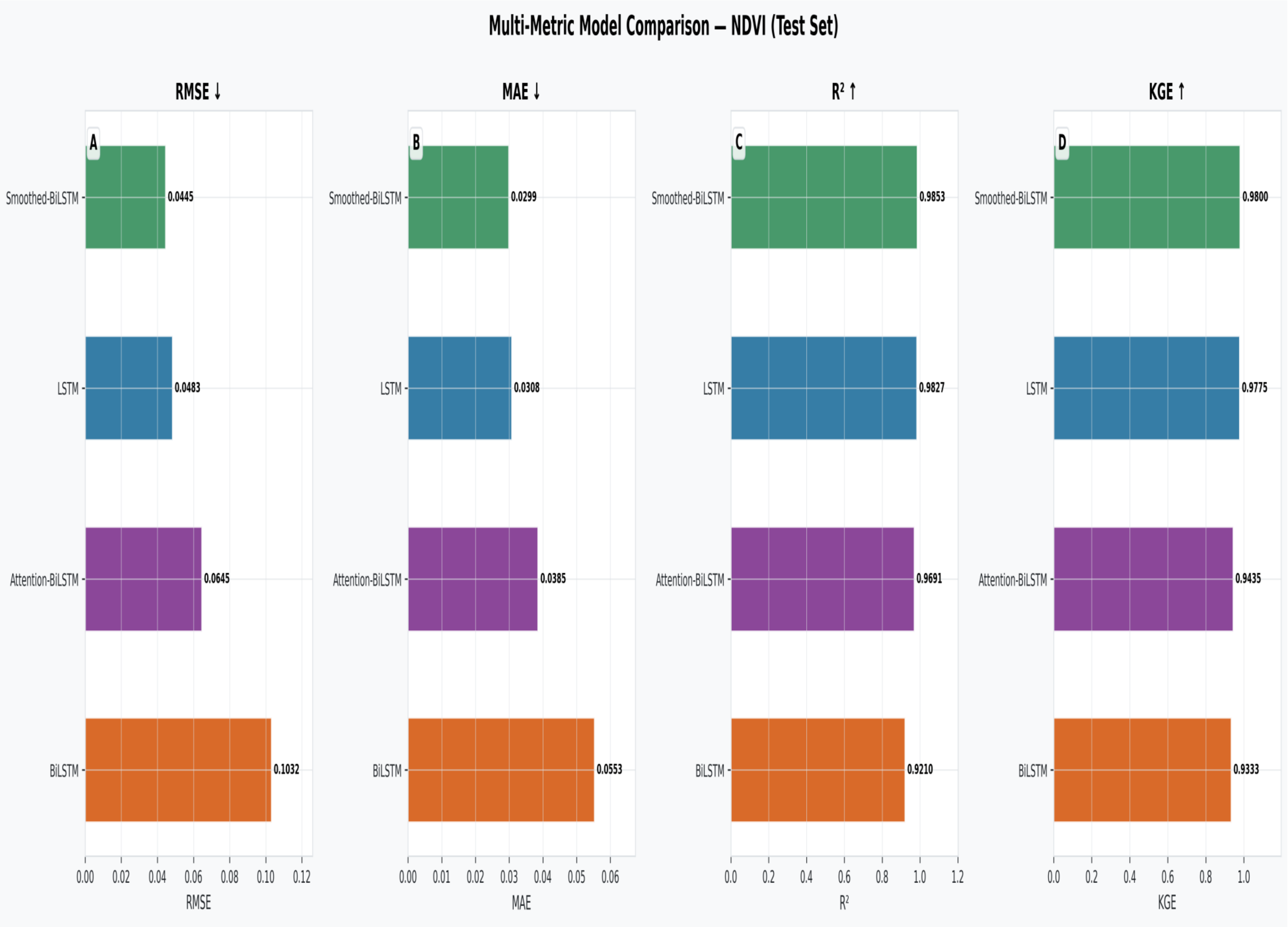


- Meteo-only models show significantly higher validation losses and unstable convergence, highlighting that meteorological variables alone are insufficient for accurate vegetation index prediction without spectral history

Data Analysis

Comparative Model Performance on the Test Set





- The LSTM and Smoothed-BiLSTM architectures outperform other models in predicting NDVI and NDMI, with the best NDVI model achieving R^2 above 0.985 and the NDMI models showing moderate skill (R^2 0.845–0.913), indicating higher predictability for NDVI than NDMI due to the moisture index's greater complexity and variability.

Data Analysis

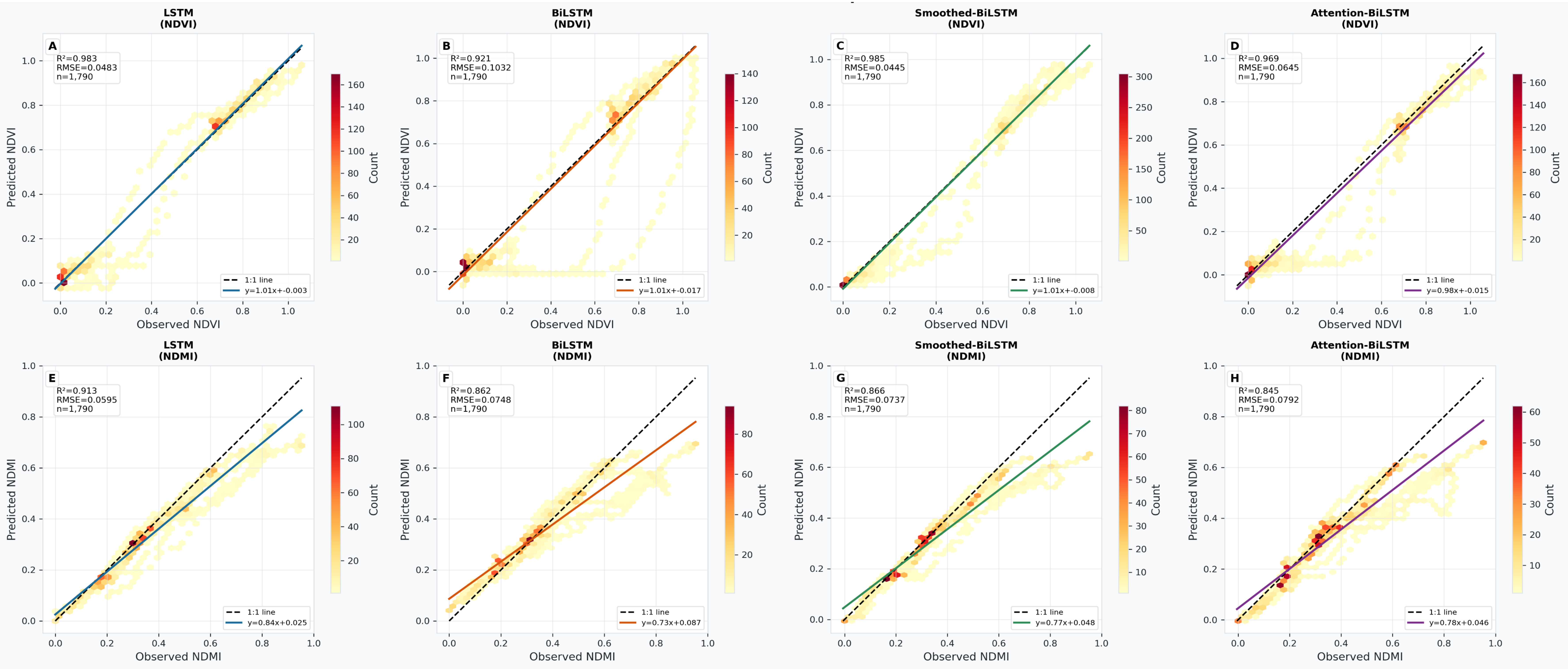
Comprehensive Evaluation Metrics Summary

Model	Target	N	RMSE	MAE	R^2	KGE	NSE	PCC	Bias
LSTM	NDVI	1790	0.04826	0.03077	0.9827	0.9775	0.9827	0.9917	0.00206
BiLSTM	NDVI	1790	0.10316	0.05526	0.921	0.9333	0.921	0.964	-0.0119
Smoothed-BiLSTM	NDVI	1790	0.04452	0.02986	0.9853	0.98	0.9853	0.9929	-0.00386
Attention-BiLSTM	NDVI	1790	0.06452	0.03853	0.9691	0.9435	0.9691	0.9865	-0.02325
LSTM	NDMI	1790	0.05953	0.03545	0.9126	0.8314	0.9126	0.9763	-0.03313
BiLSTM	NDMI	1790	0.07482	0.04248	0.862	0.7601	0.862	0.9476	-0.01229
Smoothed-BiLSTM	NDMI	1790	0.07373	0.04044	0.866	0.7747	0.866	0.9607	-0.0357
Attention-BiLSTM	NDMI	1790	0.07922	0.04716	0.8453	0.7882	0.8453	0.9448	-0.03608

- Smoothed-BiLSTM consistently achieves the best overall performance across all metrics for both NDVI and NDMI predictions, with the lowest RMSE and MAE values and the highest R^2 and KGE scores. Notably, the LSTM model also performs strongly, particularly for NDVI, indicating that input smoothing enhances prediction accuracy. Conversely, the BiLSTM and Attention-BiLSTM architectures generally underperform relative to the full-feature models, with wider residual distributions and lower correlation scores.

Data Analysis

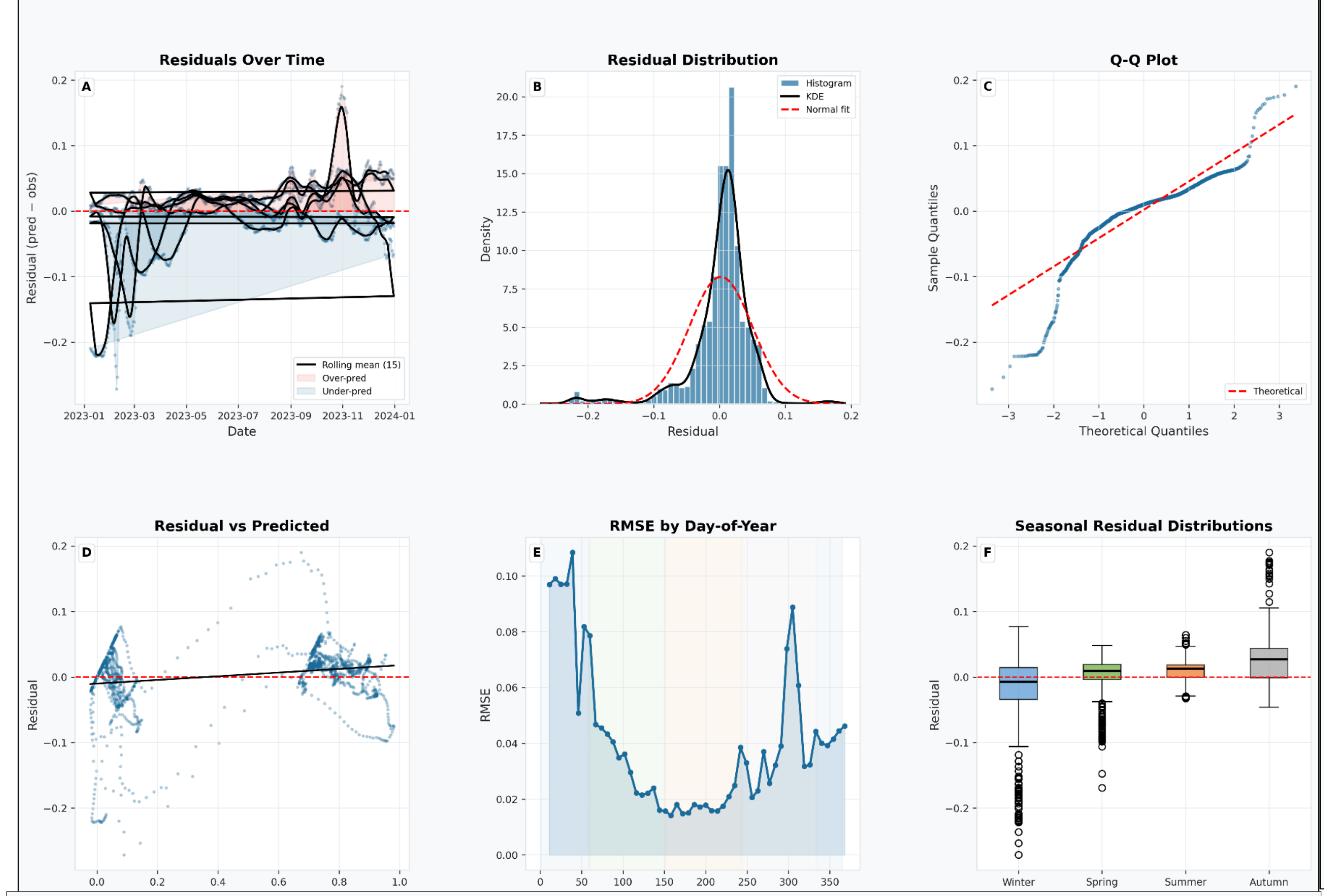
Predicted vs. Observed Density Scatter Plots

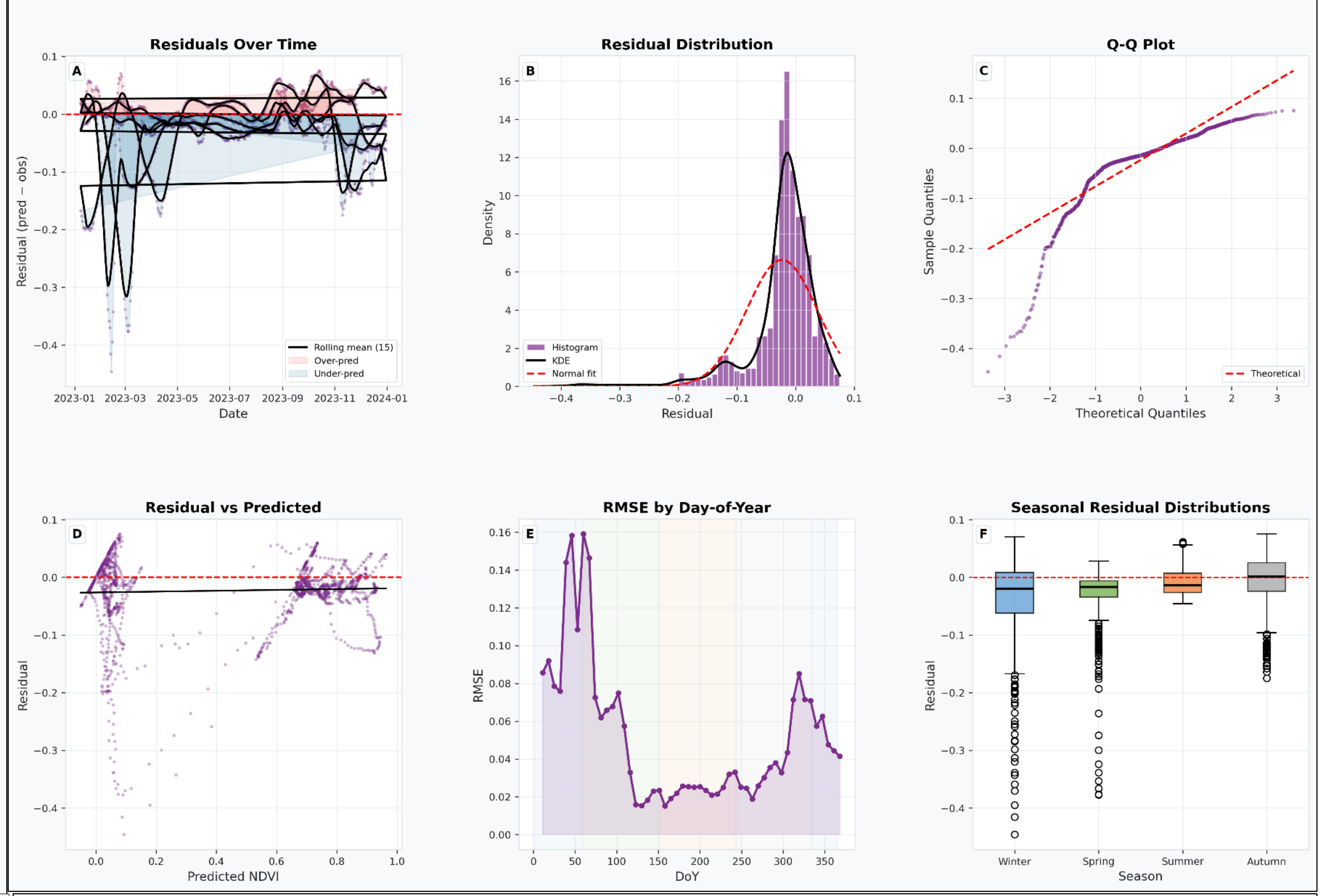


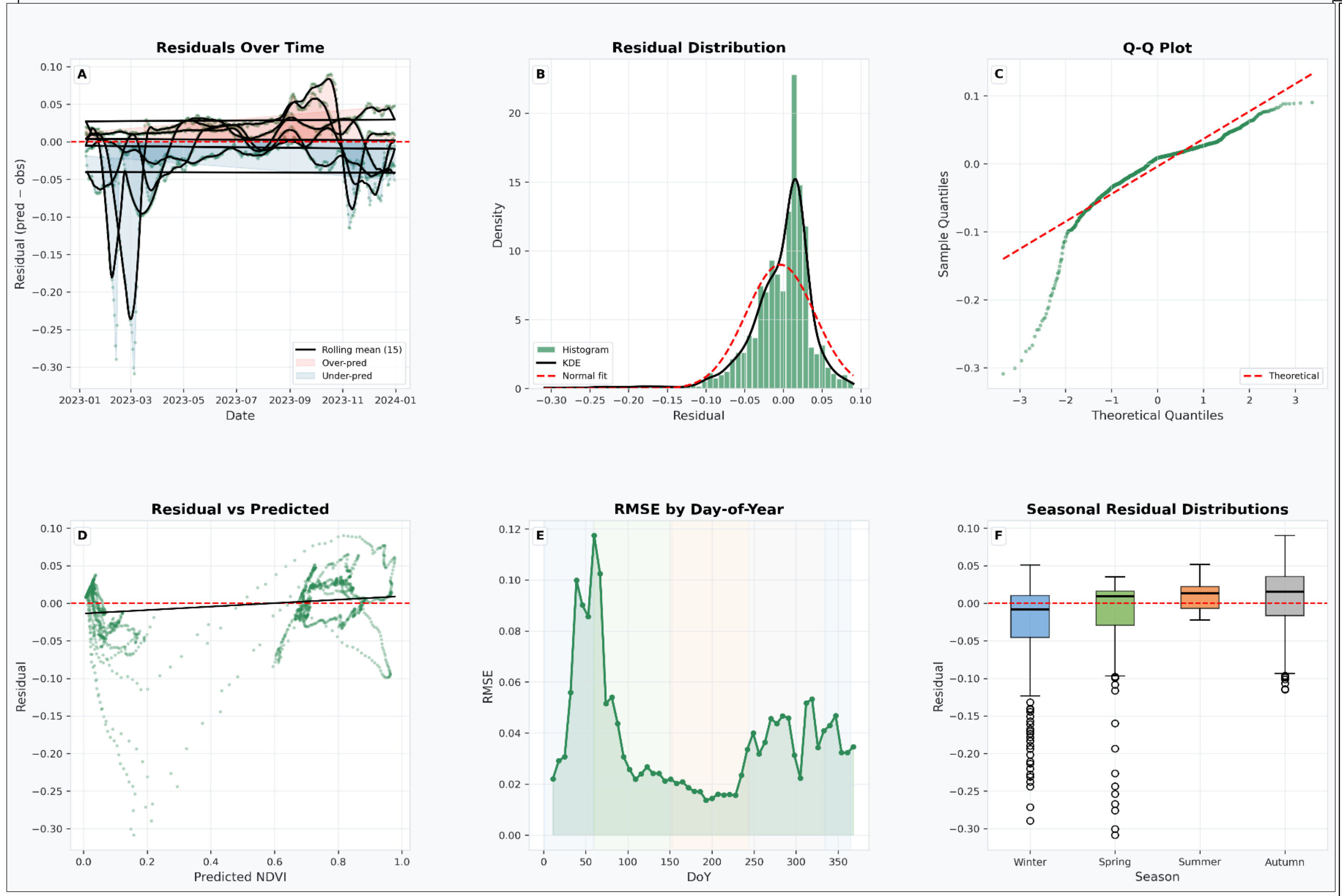
- NDVI models (Smoothed-BiLSTM and LSTM) closely follow the 1:1 line, indicating high accuracy; BiLSTM shows offset and broader errors; all NDMI models exhibit compression and bias toward mean, with moisture index predictions affected by high-frequency variability.

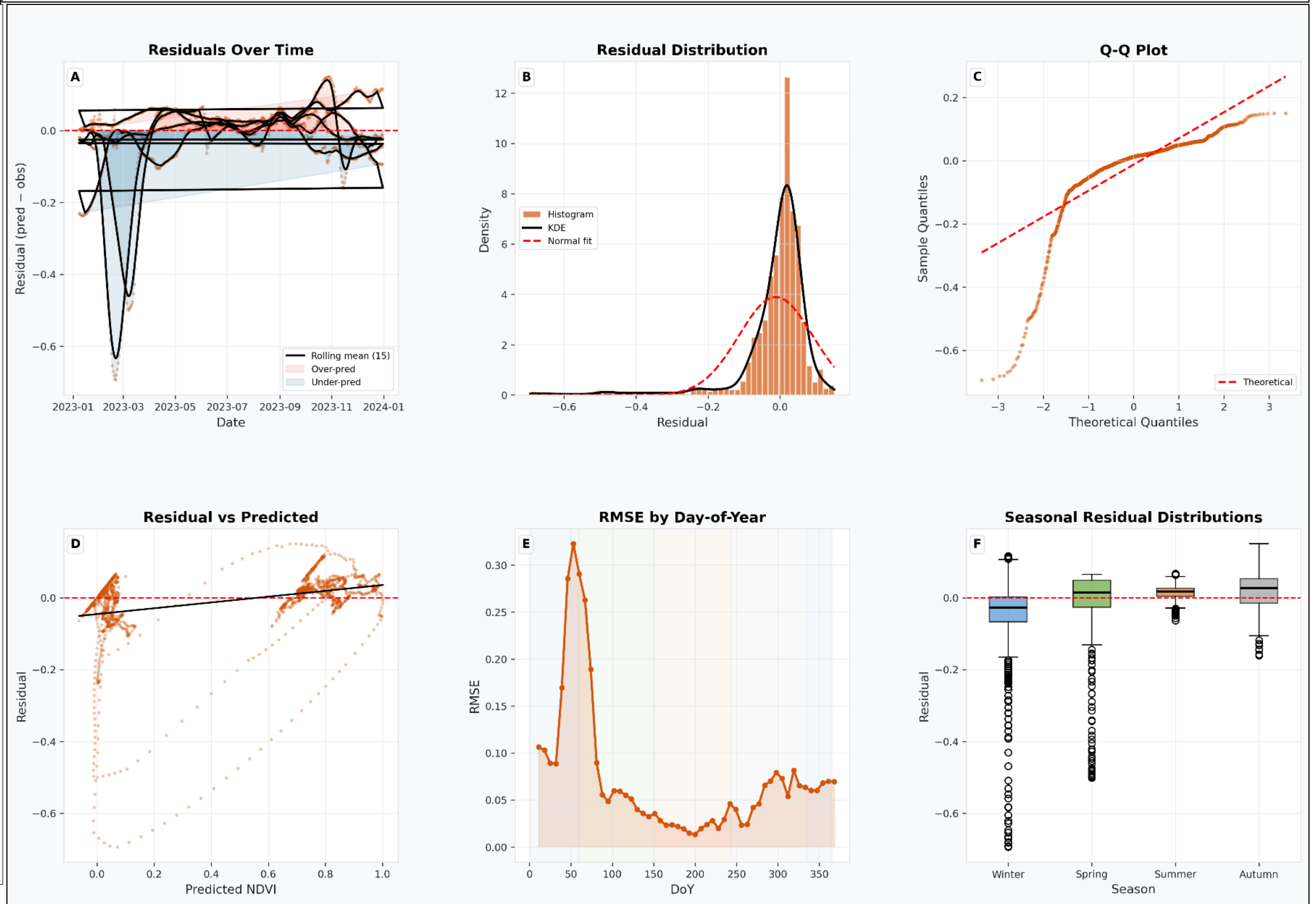
NDVI Residual Diagnostics

- The BiLSTM exhibits wider residuals, heavy skewness, and poor distributional fit, especially during phenological transitions like late winter to early spring; residuals are notably under-predicted during these periods, indicating challenges in modeling abrupt vegetation dynamics.



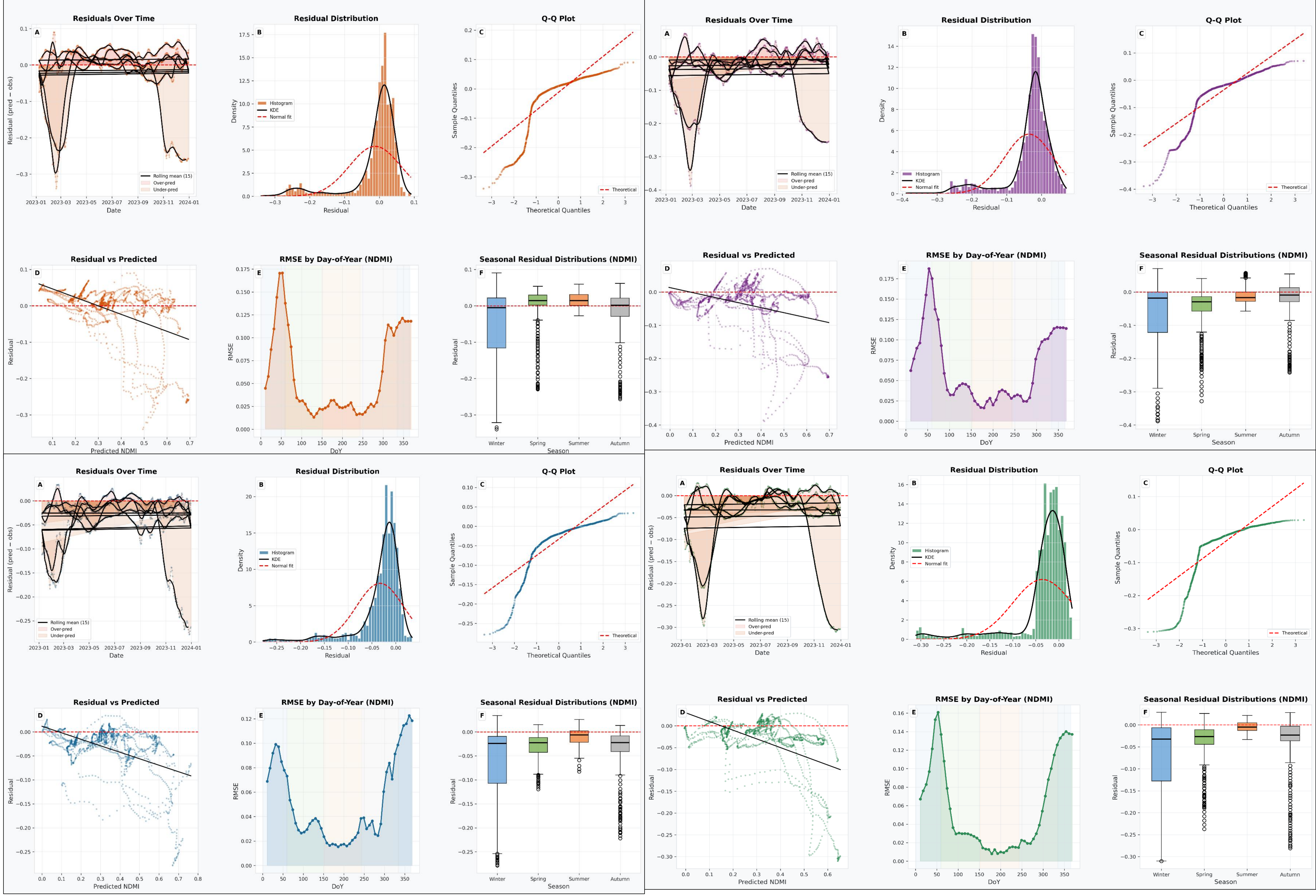






NDMI Residual Diagnostics

- All models systematically under-predict NDMI, indicating a compression bias in moisture index predictions..



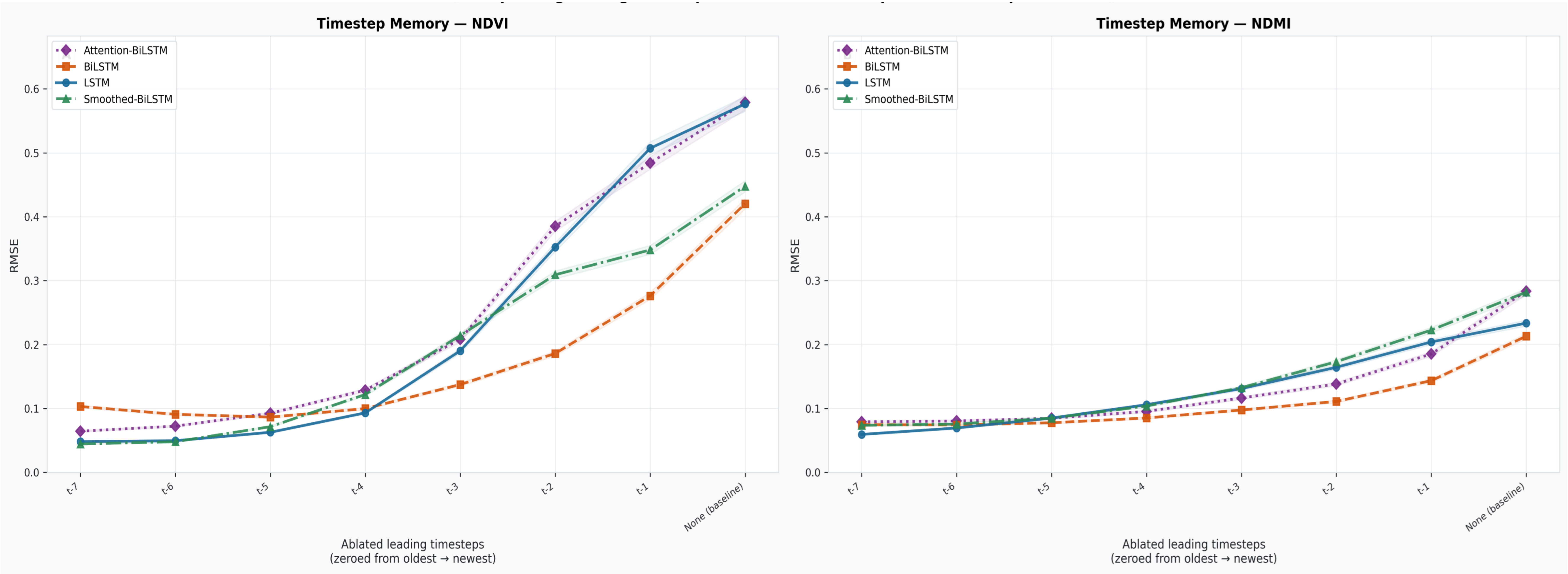
Data Analysis

Seasonal RMSE Analysis

- During seasonal analysis, the models generally perform best in summer with the lowest RMSE values (e.g., 0.0187–0.0247 for NDVI and 0.0127–0.0279 for NDMI), and worst in winter, where RMSE values are highest, such as the BiLSTM reaching 0.1615 for NDVI and 0.124 for NDMI; this pattern reflects the increased unpredictability of vegetation signals during dormant and snow-covered periods.

Data Analysis

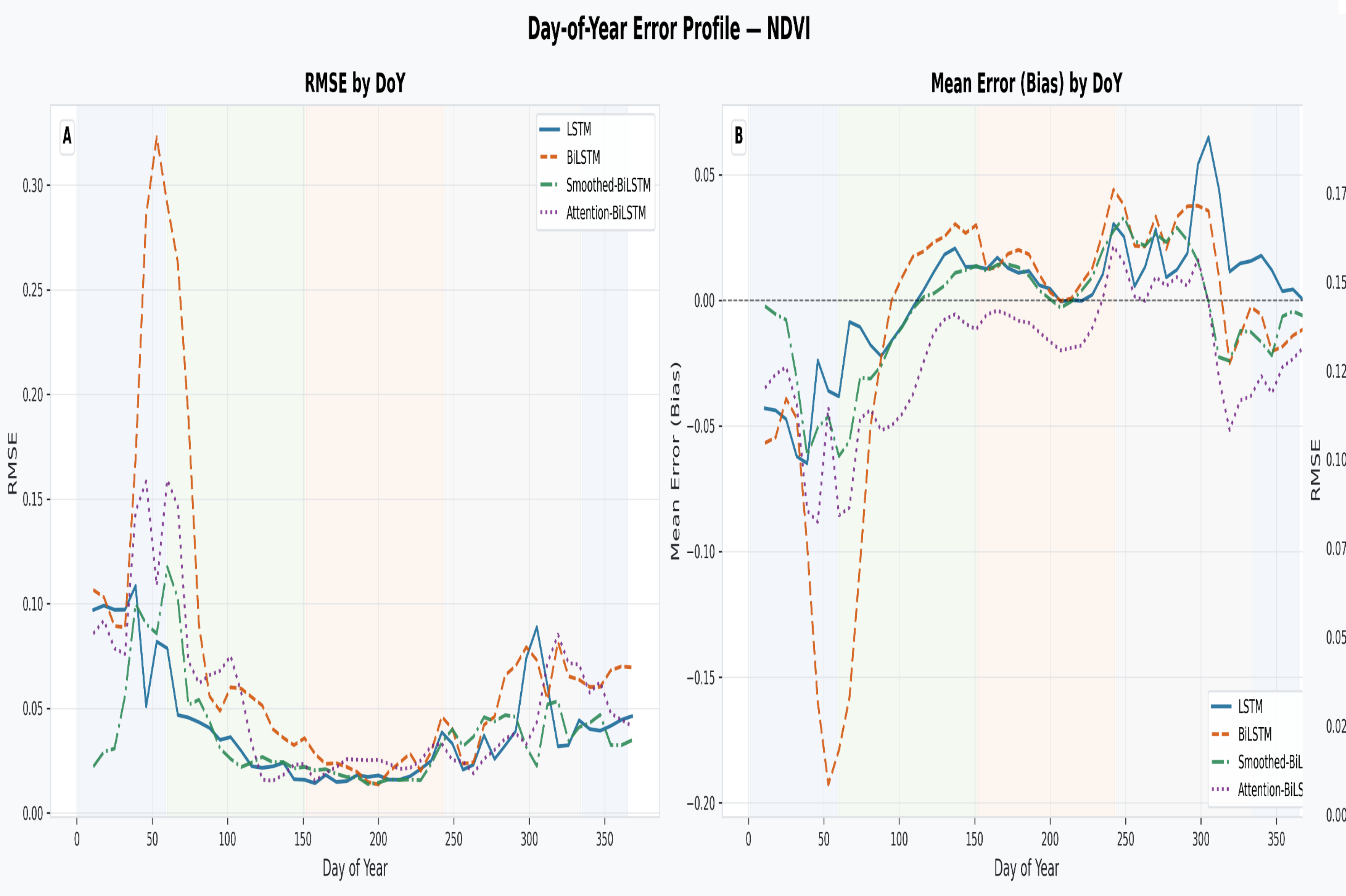
Temporal Memory Ablation Study

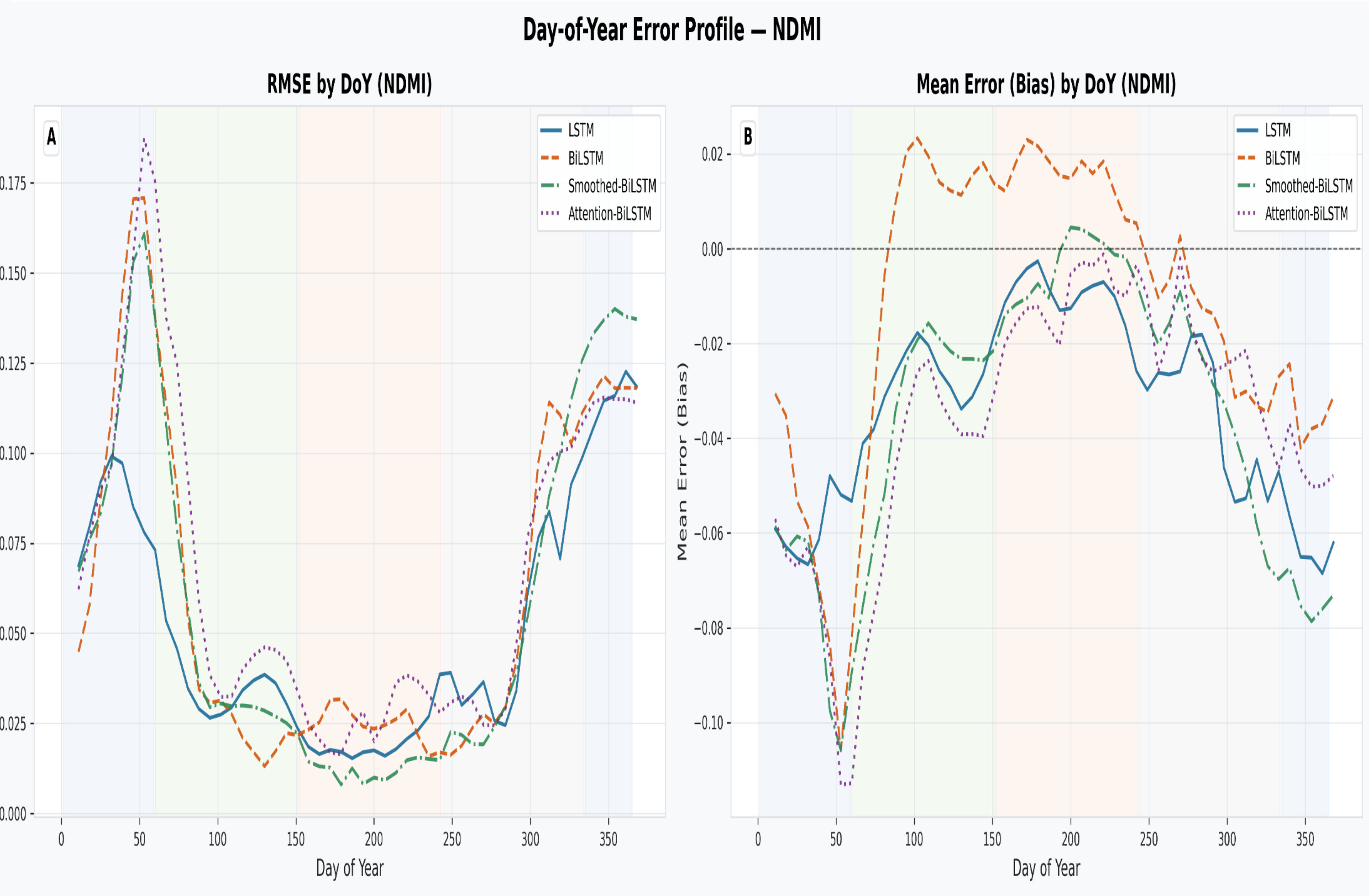


- A minimum of about four previous observations (t-4 to t-3) are needed for reliable NDVI and NDMI prediction, with RMSE increasing significantly when fewer are available, highlighting the crucial role of historical data in gap-filling accuracy.

Data Analysis

Day-of-Year Error Profiles

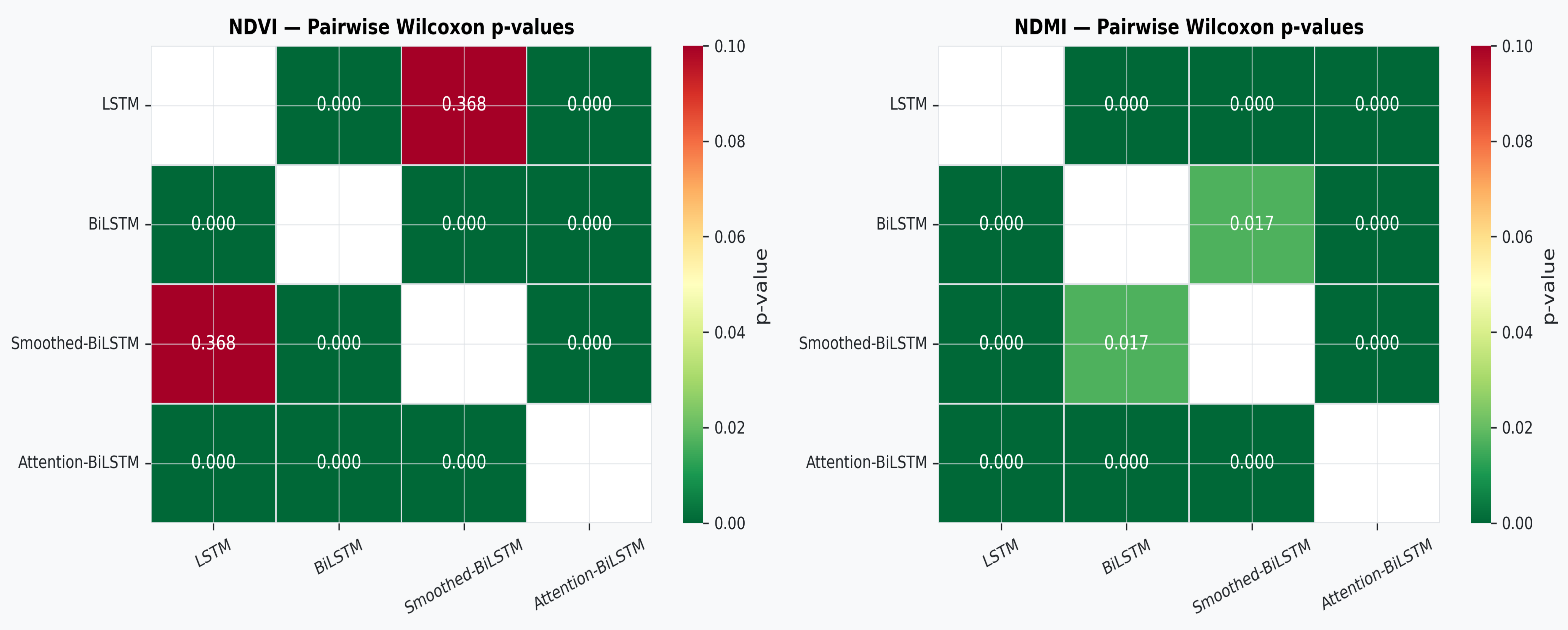




- Prediction errors peak mid-February (DoY 40-50) due to snow melt, frost, and cloud cover, causing high uncertainty in satellite-based vegetation indices.

Data Analysis

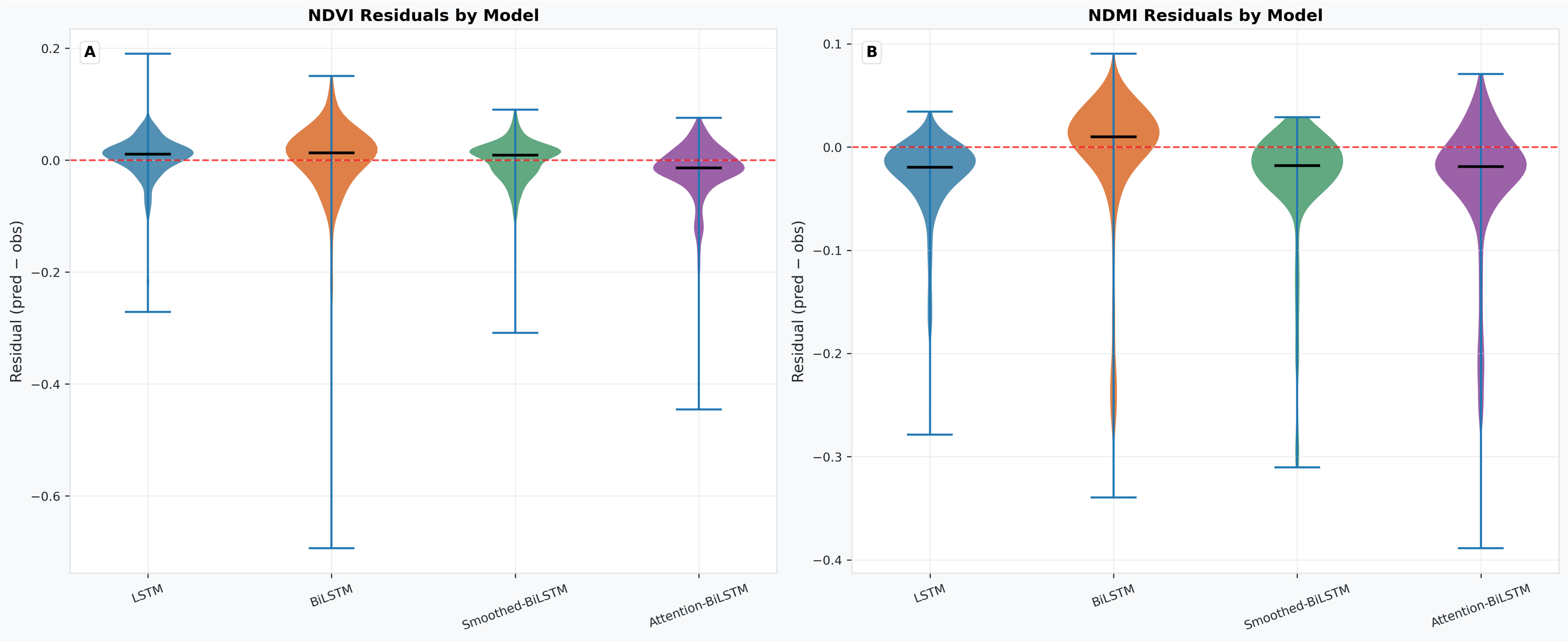
Statistical Significance Testing — Wilcoxon Signed-Rank Test



- Using pairwise Wilcoxon signed-rank tests, most model performance differences were statistically significant (p < 0.001), confirming the superior performance hierarchy of models like LSTM and Smoothed-BiLSTM, with the exception of NDVI comparison where their difference was not significant (p = 0.368).

Data Analysis

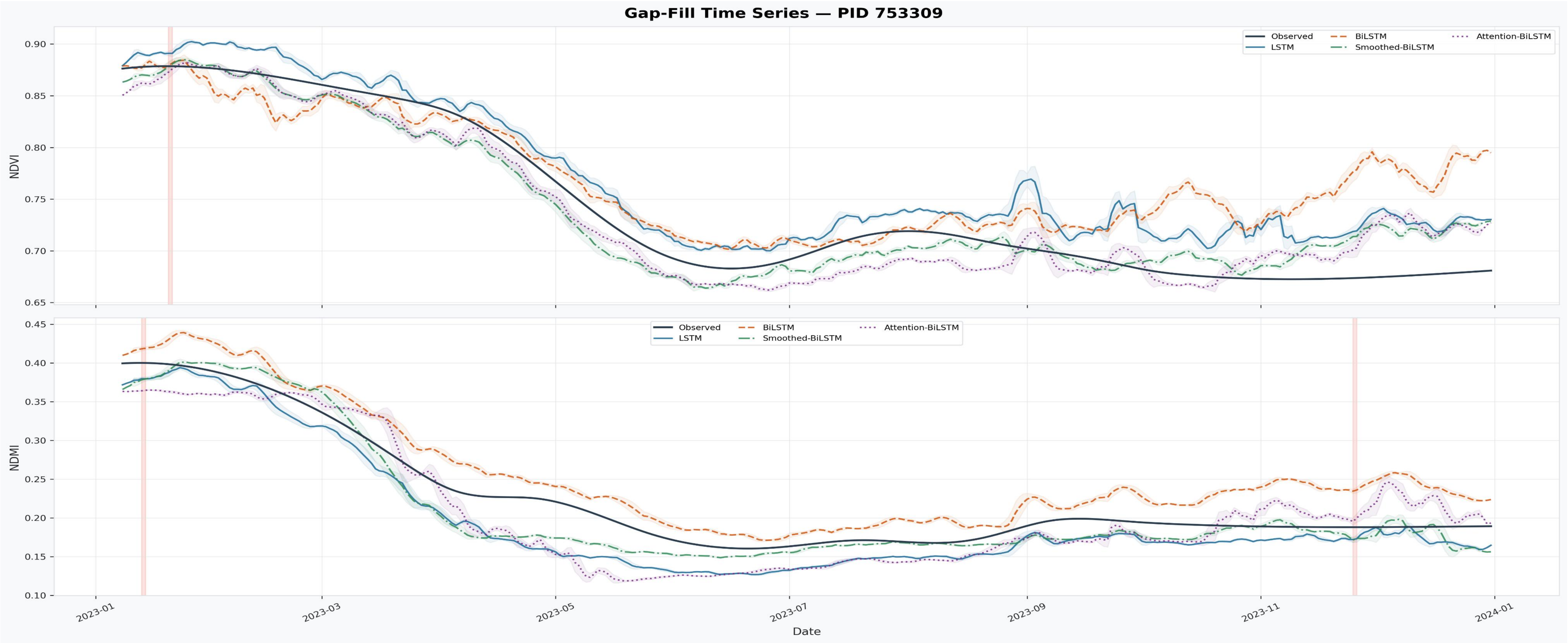
Residual Distribution Comparison — Violin Plots



- The violin plots indicate that LSTM and Smoothed-BiLSTM residuals are the most symmetric and narrowly distributed with medians near zero, demonstrating better calibration, while BiLSTM and Attention-BiLSTM exhibit wider, skewed residual distributions with systematic biases, especially underprediction for NDMI and overprediction for NDVI.

Data Analysis

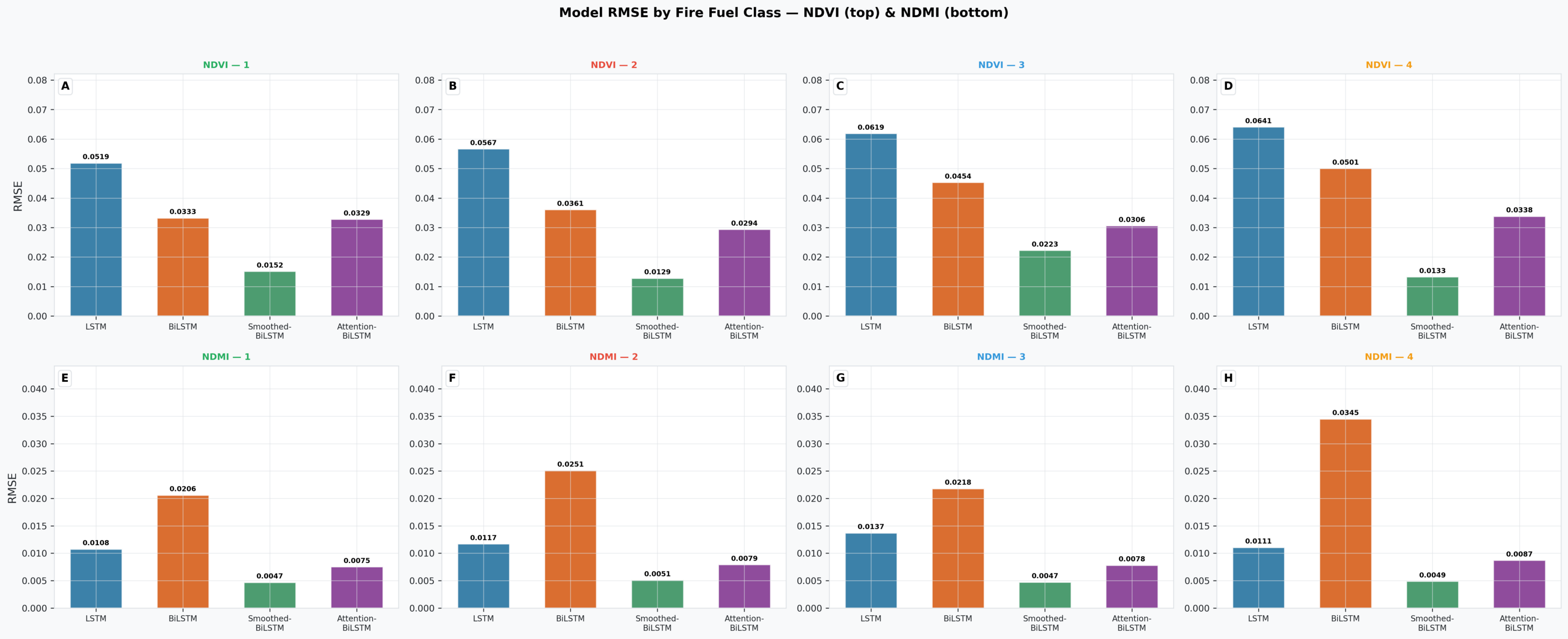
Gap-Fill Time Series Visualisation



- The gap-fill time series visualization shows that the LSTM and Smoothed-BiLSTM models closely follow observed vegetation index signals throughout the year, effectively capturing seasonal dynamics with minimal divergence. In contrast, the BiLSTM tends to over-predict during winter-spring and at year's end, while the Attention-BiLSTM exhibits under-prediction during summer and late year, highlighting their respective biases and varying suitability for operational gap-filling.

Data Analysis

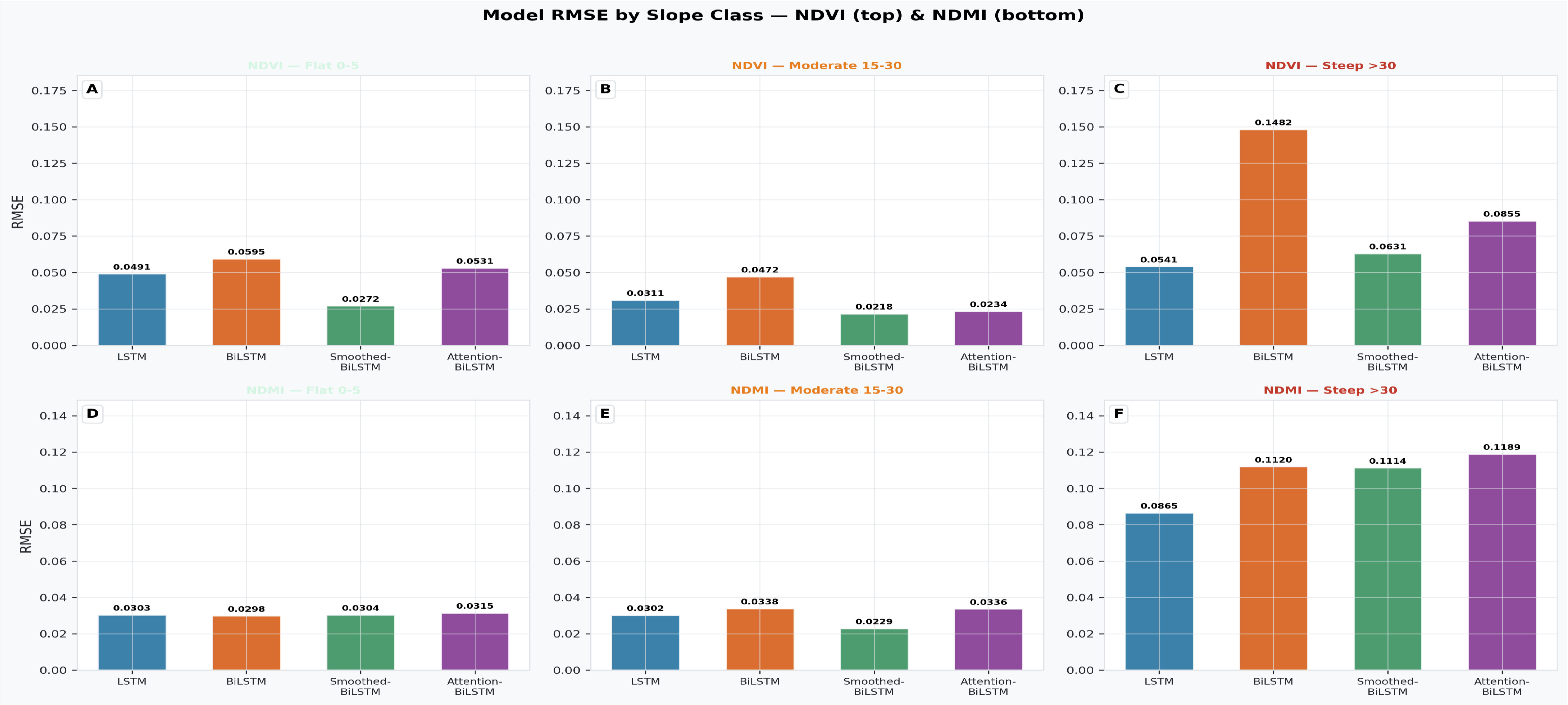
Performance by Fire Fuel Class



The study reveals that the Smoothed-BiLSTM consistently achieves the lowest RMSE across all fire fuel classes for NDVI, indicating superior accuracy in diverse vegetation density conditions. Conversely, the BiLSTM exhibits the highest RMSE, especially in dense forested areas (fuel class 3), reflecting greater prediction uncertainty in complex canopy environments.

Data Analysis

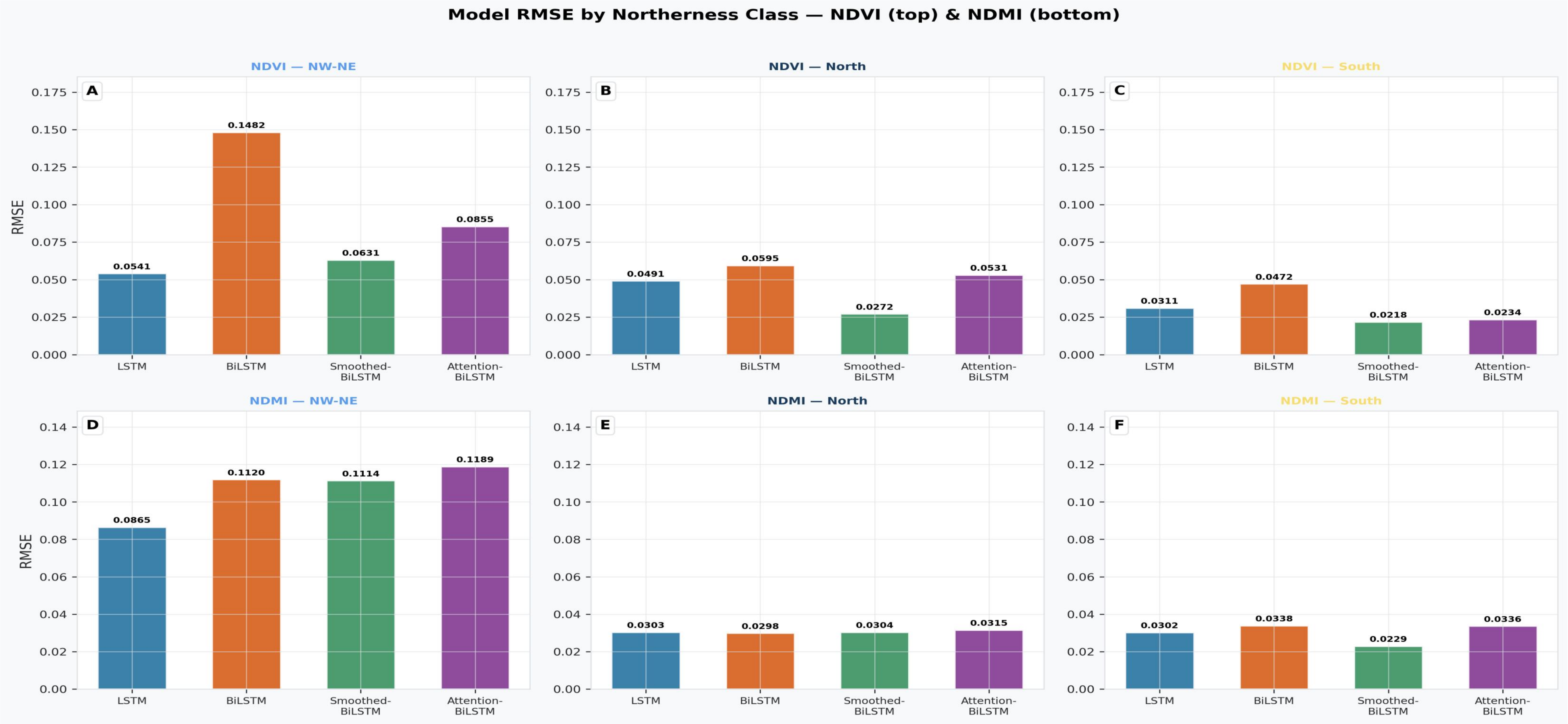
Performance by Slope Class



- Model performance for NDVI worsens with increasing slope steepness, with the BiLSTM showing significant degradation on steep terrain, while the Smoothed-BiLSTM maintains low RMSE across all slope classes, demonstrating its suitability for complex topography.

Data Analysis

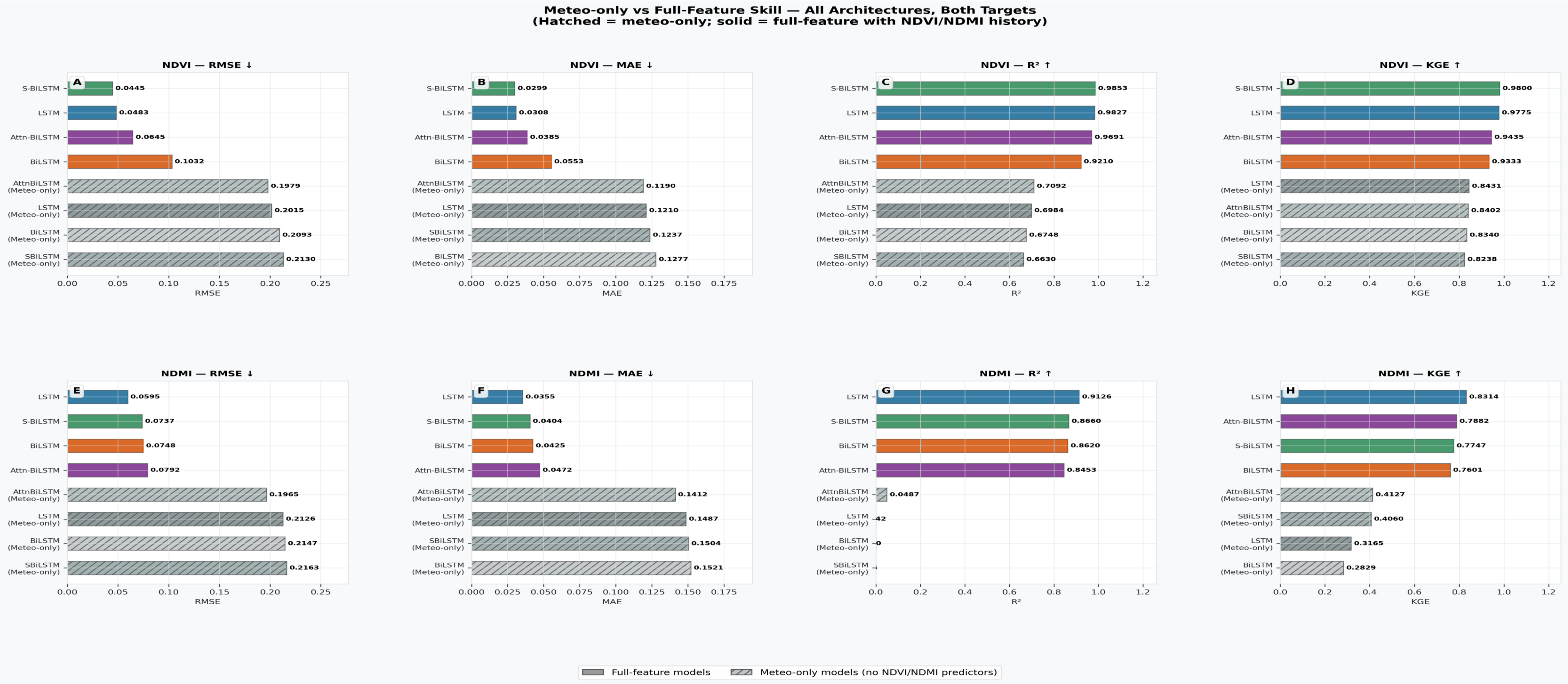
Performance by Northerness Class



- Performance stratified by aspect reveals that models, especially LSTM and Smoothed-BiLSTM, perform better on south-facing slopes, with lower RMSE values, whereas north-facing slopes exhibit higher RMSE, indicating greater prediction difficulty likely due to longer snow cover, reduced solar radiation, and vegetation activity constraints in shaded terrains.

Data Analysis

Meteo-Only vs. Full-Feature Skill Comparison



- Full-feature LSTM models significantly outperform meteorology-only models, highlighting the importance of including vegetation history for accurate gap-filling.

Discussion of Findings

- The Smoothed-BiLSTM and LSTM architectures outperform more complex models like Attention-BiLSTM and standard BiLSTM, emphasizing the importance of input preprocessing over architectural complexity.
- Input Quality vs. Architectural Complexity: Preprocessing methods such as Blackman FIR smoothing are as crucial as model complexity for improving vegetation index prediction accuracy, aligning with prior studies (Dey et al., 2024; Michishita et al., 2014).
- Performance for NDMI is lower than NDVI, reflecting the greater sensitivity of moisture indices to heterogeneity, precipitation events, and canopy structure — moisture dynamics are asymmetric and better modeled by unidirectional sequences.
- Model accuracy varies with terrain; higher errors are noted on north-facing slopes, steep terrain, and dense forests, indicating that terrain-aware strategies are necessary for operational deployment.
- A minimum of approximately 3–4 days of recent observations are required for effective gap-filling.
- Assimilation of Vegetation History: Incorporating recent vegetation index observations is indispensable; meteorological inputs alone are insufficient for reliable gap-filling, underscoring the critical role of spectral history.
- Model performance peaks during the summer growing season and declines during dormancy periods, especially winter, where snow cover, frost, and cloud cover pose significant challenges.

Conclusion

- Four LSTM architectures (Stacked LSTM, BiLSTM, Smoothed-BiLSTM, Attention-BiLSTM) were evaluated for NDVI and NDMI gap-filling over Lombardy.
- Smoothed-BiLSTM and LSTM achieved best performance; preprocessing (smoothing) is as effective as complex architectures.
- Vegetation index history is essential; meteorology alone is inadequate for reliable predictions.
- Performance varies spatially, with topography affecting accuracy; models are more reliable in flat, sunny areas.
- Minimum temporal window of 3–4 days is needed for effective gap-filling; longer gaps reduce accuracy.
- Findings inform operational systems: incorporate spectral history, apply noise filtering, and consider terrain effects for better reliability.

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